

Should an AI Agent Call a Tool, or Refuse?

3,491

Agent Tasks

33

Features

2

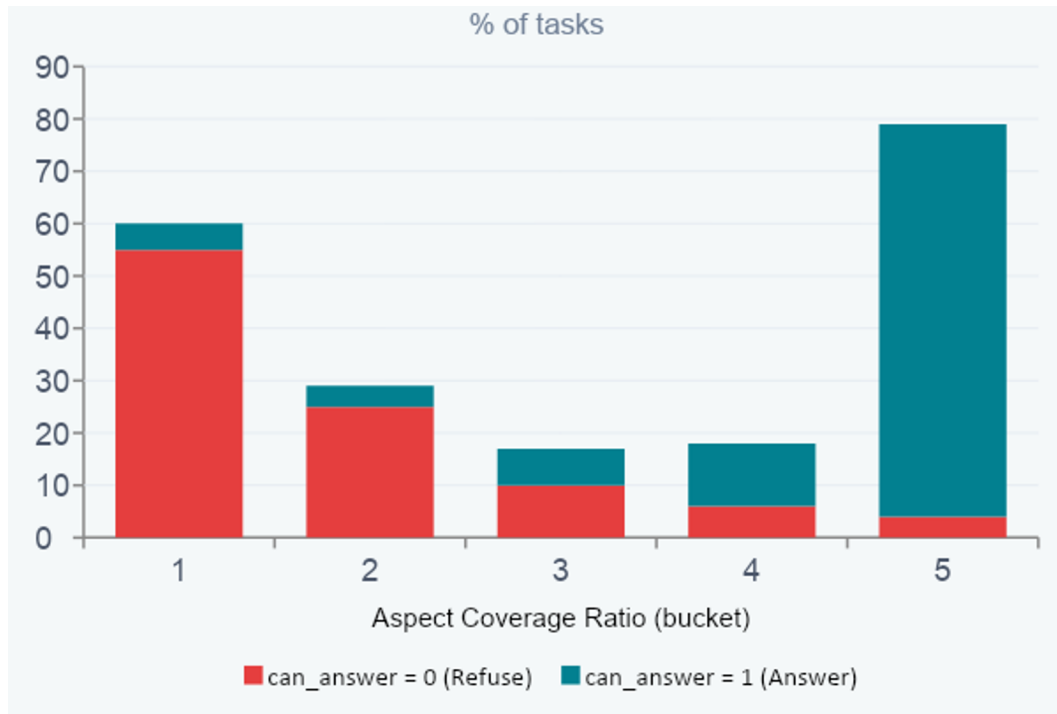
Classes

can_answer: 0 | 1

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Aspect Coverage Ratio Separates the Classes



Tasks that can be answered overwhelmingly have ratio near 1.0 — tools match the query topics.

Refusal tasks cluster at 0 — available tools are irrelevant to the query.

Strongest class separator observed in EDA, consistently top feature by importance.

Random Forest — Test Set Results

89%

Accuracy

0.8857

89%

Precision

0.8924

95%

Recall

0.9451

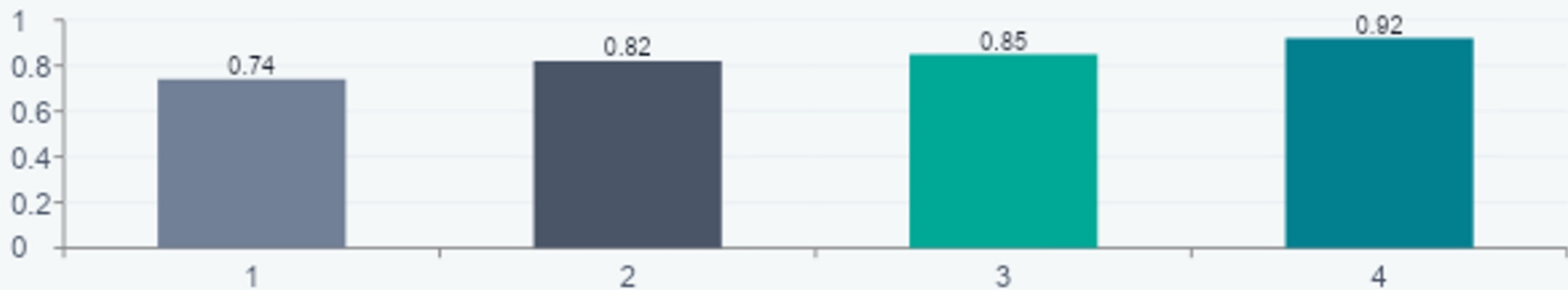
92%

F1-Score

0.9180

Hyperparameters: `n_estimators = 200` · `max_depth = 20`

Model Comparison — Validation F1



False Positives Are the Costliest Mistake

False Positive

Predict CAN answer · Actually CANNOT

- Agent calls a tool that cannot help
- May perform harmful actions (e.g. send, book, delete)
- Wastes resources and erodes user trust
- Harder to recover from in production

Most Dangerous

False Negative

Predict CANNOT answer · Actually CAN

- Agent refuses a solvable request
- User experience suffers — unhelpful AI
- No harmful side-effect
- User can rephrase and retry

Recoverable

Note: The deployed RF has precision 89% < recall 95% — it leans toward recall. A threshold adjustment can shift it toward higher precision (fewer FP) when a conservative production setting is required.

K-Means (k=4) — Four Natural Task Groups

C0**Low-Coverage Mixed**

Size	1,689
Answer Rate	54%
Asp. Cover	0.009
Top Domain	other
Complexity	low
Error Profile	TP + TN

C1**Vision Outlier**

Size	4
Answer Rate	25%
Asp. Cover	0.10
Top Domain	vision
Complexity	high
Error Profile	TN (tiny)

C2**Moderate-Match /
Complex**

Size	1,121
Answer Rate	75%
Asp. Cover	0.31
Top Domain	other
Complexity	high
Error Profile	Mostly TP

C3**High-Match / Answerable**

Size	677
Answer Rate	91%
Asp. Cover	0.85
Top Domain	math
Complexity	medium
Error Profile	Mostly TP

Dataset Limitation · Suggested Improvement

Dataset Limitation

The can_answer label is synthetically derived from benchmark construction rules — not from new human annotations. Models may learn patterns specific to the benchmark artifacts rather than truly generalising to real-world tool-use reasoning.

Suggested Improvement

Replace strict keyword-based aspect matching with embedding-based cosine similarity between the query and tool descriptions. This would capture semantic equivalences (synonyms, paraphrases) that keyword rules miss, creating far richer and more robust features.